

MARTI SOURA VAMSEEKAR

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EU Blue Card Eligible | Open to relocation: Germany, Netherlands, Ireland, Austria

PROFESSIONAL SUMMARY

Artificial Intelligence and Infrastructure Engineer with production experience in RAG / LLM pipelines, event-driven microservices, and cloud data platforms. Delivery spans fintech AI research (Bharat Alpha), EU compliance analytics (WorkforceGuard), and government-scale geospatial intelligence (Aequitas), built on Google Vertex AI, Java / Spring Boot, Python, Azure Data Factory, Databricks, Docker, Kubernetes, and GCP. M.Sc. Data Science, University of Greenwich (Merit). Microsoft Certified Azure Data Engineer Associate (DP-203). Targeting EU-based AI Engineer or Data Platform Engineer roles.

TECHNICAL SKILLS

- **AI and GenAI:** Google Vertex AI, Gemini 2.5 Flash, Google ADK, RAG (FAISS, BM25, Cross-Encoder Reranking), LangChain, LLM Agents, PaddleOCR, scikit-learn, TensorFlow
- **Cloud and Infrastructure:** Azure Data Factory, Synapse Analytics, Databricks, Data Lake Gen2, GCP Cloud Run, Docker, Kubernetes, GitHub Actions CI/CD
- **Data Engineering:** Python, PySpark, dbt, ETL/ELT pipelines, Apache Spark, DuckDB, Parquet, Pydantic, Power BI, Tableau
- **Backend and Distributed Systems:** Java 17/21, Spring Boot 3, FastAPI, Spring WebFlux, Spring Cloud Gateway, RabbitMQ, WebSockets (STOMP/SockJS), Flyway, OpenAPI, Bucket4j
- **Databases:** PostgreSQL, MongoDB, Redis, ChromaDB, Supabase, pgvector, SQLite, DuckDB
- **Frontend and Mobile:** TypeScript, React 18/19, Next.js, React Native 0.83, Flutter, Redux Toolkit, shadcn/ui, Tailwind CSS

WORK EXPERIENCE

Software Development Engineer - AI / Full-Stack

Sep 2025 - Present

Innosolv Private Limited | London, UK (Remote) | Fintech - Algorithmic trading and AI-powered equity research for NSE cash and F&O markets

- Owning end-to-end AI platform delivery for **Bharat Alpha**, an institutional equity research system for **10 NSE sectors**; Gemini 2.5 Flash enforces **10 Tier-1 screening filters** (ROE >15%, ROCE >15%, D/E <0.8, ICR >2.0, FCF >5% and 5 others) and a **100-point Tier-2 scoring model**; SSE streaming with **Supabase Auth** and Postgres-persisted conversation history.
- Architected production RAG infrastructure across **305 annual report filings for 52 Nifty 50 companies (FY2020-FY2025)**; pipeline combines **FAISS IndexFlatIP** vector index, **BM25** lexical retrieval, **reciprocal rank fusion**, and **cross-encoder reranking** into a unified **143K-chunk** pipeline with sector-aware routing across **5 verticals**; SSE-streamed responses with inline source citations.
- Designed and deployed **algorithmic trading engine** (Java 17, Spring Boot, MongoDB) with **AsyncOrderProcessor (CompletableFuture)** for non-blocking dispatch and **Bucket4j rate limiter at 10 req/s**; derivatives suite covers **Iron Condor P&L (max-profit filter >2%)**, multi-leg spread margin engine, **basket order aggregator**, and participant-wise open interest tracker.
- Built production GCP infrastructure: **WebSocket coordination + Redis caching** for real-time market data; **pgvector + Supabase** persistence layer for AI conversations; **platform observability** on GCP Cloud Run with zero-downtime deployments via **GitHub Actions CI/CD** and Docker containerisation; end-to-end system reliability owned from retrieval architecture through to LLM prompt engineering.

PROJECTS

WorkforceGuard AI - EU Pay Transparency and Workforce Intelligence [GitHub](#)

Jan 2024 - Present

Full-stack analytics | dbt + DuckDB + FastAPI + React 19 | EU Pay Transparency Directive 2023/970/EU | workforceguard-ai.vercel.app

- End-to-end analytics platform across **16 Eurostat datasets (LFS, JVS, SES), EU27, 11 NACE sectors**; **28-model dbt pipeline** computing 4 composite indices (HPI, LR, ERS, TR) with **SHA-256 hash-chained audit log** meeting EU Pay Transparency Directive 2023/970/EU requirements; CI/CD to GCP + Vercel.
- ML pipeline: **7 models, 32,769 training samples, 912K-record test set**; Random Forest: **94.7% accuracy, AUC 0.855, 99.5% recall** (CV std 0.001); evidence-bounded Gemini analyst hard-refuses on insufficient data coverage.
- Proved labour market tightness positively correlated ($r = +0.41$) with gender pay gaps across **20 EU member states** - first quantitative evidence using integrated **HPI x ERS Combined Risk Quadrant** typology.

Masova - Cloud-Native Agentic Restaurant Management Platform [GitHub](#)

Feb 2023 - Present

8 autonomous AI agents | 6 Spring Boot 3 / Java 21 microservices | GCP Cloud Run | GDPR Art. 17 | EU VAT (12 countries)

- Deployed **8 Google ADK 1.25 / Gemini 2.0 Flash autonomous agents** with tool-use orchestration and Redis session management; agents autonomously handle **demand forecasting, inventory reorder, churn prevention, and customer review drafting** end-to-end - a fully agentic AI operations layer with InMemory fallback.
- Event-driven backend: **6 Spring Boot 3 / Java 21 microservices**, 399 API endpoints, **11-state order state machine**, dual-write PostgreSQL + MongoDB, **3 RabbitMQ topic exchanges with DLQ**; PIN lookup **200ms to 5ms (97.5%)**; EU VAT engine (TSE, NF525, FDM, RT, NTCA) covering **12 countries**.
- React Native 0.83 staff app with **offline queue buffering 1,000 actions / 30s sync**, native Android background GPS; **3 notification channels** (Brevo, Twilio, Firebase); GDPR Art. 17 compliant with **2-yr customer / 7-yr PCI retention** workflows.

Aequitas - Public Transport Equity Intelligence Platform [GitHub](#)

Oct 2021 - Present

M.Sc. Dissertation to Production | FastAPI + DuckDB + FAISS RAG + React/Vite | Applicable to EU, UK, Australia, NZ, USA transport networks

- 7-stage validated pipeline processing **1.75M GTFS trips, 13,099 routes, 274,719 stops, 33,755 LSOAs** (56.5M population); **103 quality checks, 0 failures**, spatial join at **99.9993% accuracy**; framework designed for GTFS-compliant networks across EU, UK, Australia, New Zealand, and USA.
- ML stack: Random Forest (R2 0.472, top SHAP: noacar_pct), HDBSCAN, Isolation Forest; **25FCA accessibility scoring** within 400m catchment; DfT WebTAG v2.03fc BCR at 3.5% HM Treasury discount rate; equity outcomes: **Gini 0.5741, Palma 5.702, 4,245 zero-stop LSOAs, 5,189 evening-isolated (15.4%), 612 triple-deprived communities**.
- FAISS RAG chatbot (all-MiniLM-L6-v2 + Gemini 2.5 Flash) powers a **policy Q&A dashboard** across **51 analytical sections, 30 Jinja2 templates**, 8 policy dimensions; Bus Services Act 2025 franchising readiness; production-ready for DfT and Local Transport Authorities.

- AI system registry with automated **EU AI Act risk classification**, cited-evidence RAG (FAISS + pgvector HNSW), eval gate simulation delivering **PASS / REVIEW / BLOCKED** release decisions and **HMAC-SHA-256 signed audit event stream**.
- Spring Boot 3 backend: **Flyway migrations V1-V6**, select-for-update-skip-locked eval worker, **multi-tenant isolation** via scoped JPA + TenantContextFilter; Next.js 14 dashboard with **interactive DAG lineage graph** and Evidence Pack JSON export for EU AI Act documentation.
- Evidence Pack JSON generates **audit-ready EU AI Act technical documentation** at every model release cycle; RBAC + SSO roadmap; designed for **regulated AI teams** needing defensible compliance artefacts with full data lineage traceability.

RESEARCH AND PUBLICATION

Why Tight Labour Markets Do Not Close Gender Pay Gaps: Evidence from a 20-Country Eurostat Panel

Marti, S. V. (2026) | [SSRN](#) | [Zenodo](#) | ORCID: 0009-0005-4884-1292

- Applied panel data econometrics + ML across **20-country Eurostat dataset (2019-2024, 11 NACE sectors)**; found $r = +0.41$ positive correlation between labour market tightness and gender pay gaps - contradicts competitive equalisation theory; introduced **Combined Risk Quadrant (HPI x ERS)** as first integrated tightness-equity typology in literature.
- Constructed **4 composite panel indices** (HPI, LR, ERS, TR) via PCA-informed weighting across **11 NACE sectors** and **5 years of longitudinal data**; applied fixed-effects regression, Hausman specification test, and heteroskedasticity-robust SEs - demonstrating production-grade data science applied to EU labour policy.

CERTIFICATIONS

- **Microsoft Certified: Azure Data Engineer Associate (DP-203)** | Microsoft | March 2025 | Credential ID: CEA6999835BAD6E

EDUCATION

M.Sc. Data Science - Merit

Jan 2021 - Apr 2022

University of Greenwich | London, United Kingdom

- Dissertation: UK Bus Data Analysis (Aequitas) | Modules: **Big Data, Applied ML, Data Visualisation, Cloud Grids and Virtualisation**

B.Tech. Electronics and Communication Engineering - 8.3 CGPA

Jun 2016 - Mar 2020

GITAM Institute of Science and Technology | Visakhapatnam, India

- Dissertation: Hyperspectral Image Analysis with Deep Learning | Modules: **Data Structures and Algorithms, Digital Signal Processing, Computer Networks**